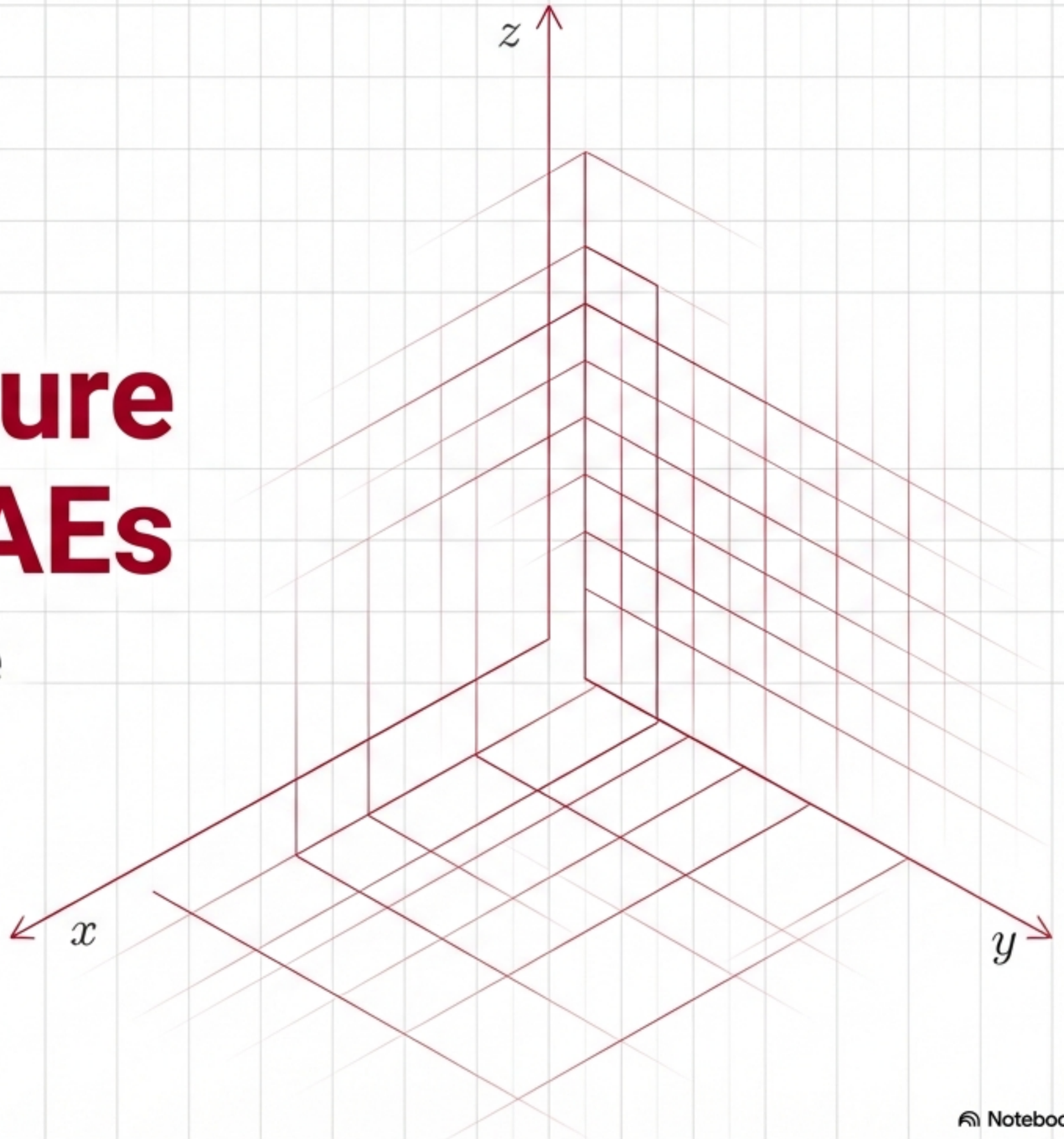
Geometric Feature Invariance in SAEs

A Framework for Transferable Mechanistic Interpretability



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The Representation Gap – Statistical vs. Semantic Hierarchies

	Current Synthetic Frameworks	New Compositional Framework
Vector Relationship	Independent random directions. $\boldsymbol{d}_i = \frac{\mathbf{s}_i}{\|\mathbf{g}_i\|_2}$ where $\mathbf{g}_i \sim \mathcal{N}(0, \boldsymbol{I}_{780})$	Literal geometric overlap (cosine similarity).
Hierarchy Enforcement	Statistical triggers. $c_{\text{child}} \leftarrow c_{\text{child}} + 1[\alpha_{\text{parent}} > 0]$	Embeds parent vector components directly into child representations.
Feature Overlap	Unstructured superposition. Forces N=16,384 into D=750 ($p_{\text{l1um}} \approx 0.15$).	Structured, semantic superposition reflecting concept taxonomy.
Metric of Success	Purely statistical decomposition.	Genuine semantic compositional understanding.

Key Insight: Current benchmarks test **statistical** dependencies, but **fail** to evaluate if SAEs recover **semantically meaningful compositional structures** where a child concept **literally contains components of its parent**.

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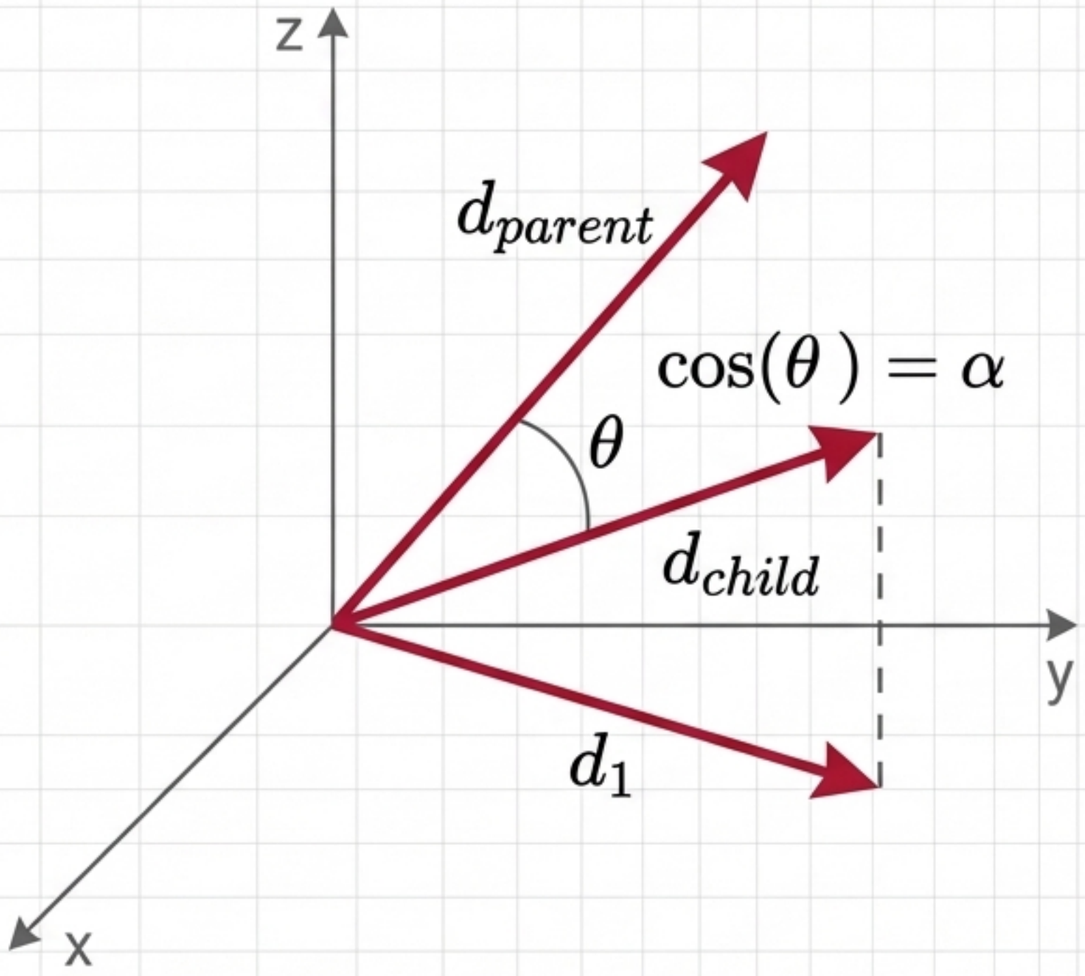

Compositional Equation:

$$d_{child} = \alpha \cdot d_{parent} + \beta \cdot d_1$$

Gram-Schmidt Orthogonalization:

$$d_1 = v - (v \cdot d_{parent}) d_{parent}$$

Geometric Relatedness:

Technical Context	\α dictates the weight of semantic inheritance, while β controls the magnitude of orthogonal specialization into a unique subspace.

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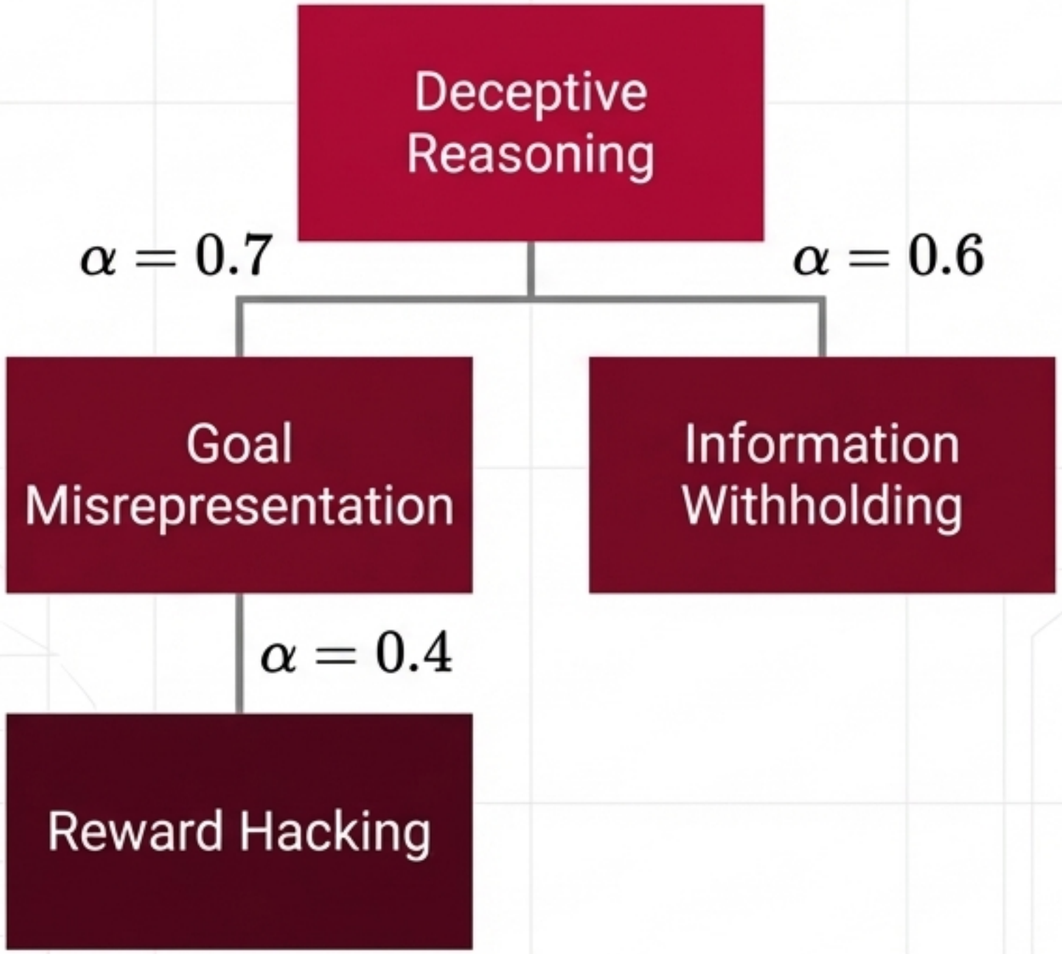
Pipeline Implementation – LLM-Generated Misalignment Hierarchies

Step 1: LLMs generate a concept taxonomy of 10,884 hierarchically constrained features (128 root trees, max depth 3, branching factor 4), assigning explicit α values for semantic similarity.



Step 2: β coefficients are derived to satisfy α constraints, constraints, and child vectors are recursively constructed. Grandchildren (e.g., Reward Hacking) transitively inherit geometric structure from both parent and root.

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Transferable AI Safety Validation

Stage 1

Statistical Synthetic Baselines

Validating SAE capabilities on complex statistical structures.

Proving competence in handling $O(1/\sqrt{D})$ superposition interference.

Stage 2

Semantic Grounding Validation

Testing across model families.

Validating that interventions ablating parent features (d_{parent}) successfully insert the reconstruction of dependent child features.

Safety Implications: Geometric feature similarity is both necessary and sufficient for interpretability transfer. By establishing structural invariants, we enable the rapid safety evaluation of fine-tuned variants and new model families $o(n = (\alpha al), \sqrt{N} - d_{parent})$ without restarting O(*N*) interpretability analysis or retraining SAEs from scratch.

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